IT Architecture

Computing and Mathematics

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

IT Architecture (A13387)

Short Title: IT Architecture
Faculty Science and Computing
Department Computing and Mathematics
Cluster Software and Web Development
Author SDROHAN
Credits 10 **Level:** Postgraduate

Description of Module / Aims

This module will provide a well-balanced understanding of how the different IT Architecture Types, Frameworks and viewpoints are essential in the successful design, deployment and management of modern Enterprise Software Systems.

Programmes

Indicative Content

- Introduction to IT Architecture
- The varying roles of the Architect: Business Perspective, IT Perspective, Leadership, Project Sponsor - Control Scope
- Types of IT Architecture: Enterprise Architecture, Business Architecture, Application Architecture, Information/Data Architecture, Security Architecture
- Architecture Frameworks: RUP (Rational Unified Process), TOGAF (The Open Group Architecture Framework)
- Zachman, IEEE 1471 (ANSI/IEEE 1471-2000, Recommended Practice for Architecture Description of Software-Intensive Systems), RM-ODP (Reference Model of Open Distributed Processing)
- Architecture Viewpoints: The need for many different system viewpoints: Functional, Information, Concurrency, Development, Deployment, Operational, Security
- Architecture Perspectives: Security, Performance and Scalability, Availability/Disaster Recovery, Hosting/Infrastructure (Clustering/Replication)
- Defining the Architecture: Modelling Languages (ADL/UML etc), Architecture Definition Documentation, Architecture Validation (scenarios, presentations, prototypes, etc)
- Architecture Controls: IT Governance, Architecture Libraries (Inventory/Lifecycles), Technology Selection, Systems/Application Roadmaps, Data Protection and Privacy Legislation
- Cloud Computing: Data Center Architectures, SOA and Cloud Computing
- Security: Security Foundations, Security Services & Patterns, Infrastructure Security, Application Security

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Evaluate the role IT Architecture has in Enterprise Software Systems.
2. Assemble an IT architecture using various documentation approaches and architectural description languages.
3. Compare and contrast different Architecture Frameworks.
4. Critique the importance of IT Architecture in an Enterprise Software System.
5. Determine and assess future trends in IT Architecture research and practice.

Learning and Teaching Methods

- This module will be presented by a combination of lectures and computer-based practicals whilst capitalising on a web-enhanced learning environment.
- The lectures will be used to introduce new topics and their related concepts.
- A cooperative learning/peer tutoring (i.e. problem solving / class discussion) approach will be adopted during the sessions.
- Self-directed learning will be encouraged throughout the duration of the module.

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	50%	1,2,3,4,5
Continuous Assessment	50%	
Assignment	50%	2,3,4

Assessment Criteria

<40%: Inability to recognise and use any architectural constraints, styles or patterns.
40%-59%: Able to recognise and use an architectural description language to present an architectural model and a range of viewpoints.
60%-69%: All of the above + be able to map an architecture to multiple architectural styles. Be able to reason with contrasting alternatives.
70%-100%: All of the above to an excellent level. Demonstrates advanced utilisation of patterns and architectural thinking.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	24
Practical	24	24
Independent Learning	222	222

Supplementary Material

- Brooks, F. _The Mythical Man-Month: Essays on Software Engineering_. New York: Addison Wesley, 1975.
- Daigneau, R. _Service Design Patterns: Fundamental Design Solutions for SOAP/WSDL and RESTful Web Services_. New York: Addison Wesley, 2014.
- De Marco, T. _Peopleware:_ _Productive Projects and Teams_. New York: Addison Wesley, 2013.
- Hohmann, L. _Beyond Software Architecture: Creating and Sustaining Winning Solutions_. New York: Addison Wesley, 2003.
- Hohpe, G. _Enterprise Integration Patterns: Designing, Building, and Depl__oying Messaging Solutions_. New York: Addison Wesley, 2003.

Requested Resources

- COMPUTER LAB: BYOD Lab